



MINING DEALER BEST PRACTICE

C175 Main Oil Gallery - D&A Debris Protection

Maintenance and
Repair

Application

Component Life
Management

Component
Renewal (CRC)

MARC
Management

C175 Main Oil Gallery - D&A Debris Protection	0
1.0 Introduction	1
2.0 Best Practice Description.....	1
3.0 Implementation Steps	2
4.0 Benefits.....	2
5.0 Resources Required	2
6.0 Supporting Attachments / References	2
7.0 Related Best Practices	2
8.0 Acknowledgements.....	2

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1.0 Introduction

Removing cylinder head(s) when doing a partial repair in the CRC can allow debris to fall into the main oil gallery through the oil supply port. Likewise, camshaft bearing installation during an engine overhaul will generate small metal shavings that will fall into the main oil gallery. This Best Practice illustrates a simple method of catching any contaminants so they can be removed.

2.0 Best Practice Description

In this example, 75mm PVC tubing has been cut in half lengthwise, and cut to block length, and inserted into the main oil gallery of a C175 engine. As the camshaft bearings are pressed into the block, any small metal shavings that are generated during installation will fall into the main gallery and be caught by the tube (tube has been slid out partly for illustration purposes).



Here is a close-up of bearing filings that were caught in the tube from one bearing.



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C175 Main Oil Gallery - D&A Debris Protection	DATE 5/12/2016	CHG NO 00	NUMBER 1605-4.01-1432

The tube can also be used during a partial repair during cylinder head(s) removal and installation by installing it into the oil gallery through the engine front housing (tube would need to be longer to take into account the width of the front housing).

The tube should be thoroughly cleaned before and after each use, and stored in a protected manner.

3.0 Implementation Steps

Obtain, cut, and de-burr the proper diameter PVC tubing, and implement a storage method.

4.0 Benefits

Maintains a clean main oil gallery prior to engine start-up.

5.0 Resources Required

Very minimal as described above.

6.0 Supporting Attachments / References

7.0 Related Best Practices

8.0 Acknowledgements

This Best Practice was developed at Finning CRC in Antofagasta – Hector Fuentes.

The photos were taken during an engine rebuild at Matco CRC in Hermosillo.

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C175 Main Oil Gallery - D&A Debris Protection	DATE 5/12/2016	CHG NO 00	NUMBER 1605-4.01-1432
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